

Quantum-Resistant Identity and Autonomous Decision Integrity: A PQC-Enabled Model Context Protocol

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Abstract—As autonomous AI agents transition from isolated tools to a globally interconnected ecosystem via the Model Context Protocol (MCP), securing their identity and the integrity of their decision-making processes against quantum-era threats is paramount. This paper proposes a Post-Quantum Cryptographic (PQC) framework for AI agents, centered on the *PQC-Audit-Chain*—a mechanism that cryptographically binds agent reasoning, tool execution, and observations using NIST-standardized ML-DSA (FIPS 204). We demonstrate how PQC signatures provide long-term non-repudiation for agentic workflows, protecting against retroactive tampering by future quantum adversaries. Through empirical analysis, we show that while PQC introduces significantly larger signatures, the integration of a hybrid identity model and efficient context management ensures that quantum-safe sovereignty is achievable with negligible impact on reasoning performance. Our work provides a reference implementation for securing the “Chain of Thought” as a verifiable, immutable audit trail in the post-quantum era.

Index Terms—Post-Quantum Cryptography, AI Agents, Model Context Protocol, ML-DSA, Zero Trust, AI Governance, Audit Chain.

I. INTRODUCTION

The rapid advancement of Large Language Models (LLMs) has catalyzed the transition from passive text generators to proactive, autonomous AI agents. These agents are characterized by their ability to perform multi-step reasoning, plan complex trajectories, and interact with the physical and digital world through tool execution [7]. As agents move beyond isolated sandboxes into enterprise environments, the need for standardized communication has led to the emergence of the Model Context Protocol (MCP) [6]. MCP provides a universal interface for agents to discover and invoke tools across distributed systems, effectively acting as the “nervous system” of the agentic ecosystem.

In a Zero Trust architecture, such as the `llm-cli` framework, securing this nervous system is paramount. Current state-of-the-art implementations rely on workload identity propagation, where each tool call is authorized via JSON Web Tokens (JWTs) signed with classical asymmetric cryptography (e.g., RS256 or ES256). These tokens bind the agent’s intent to a specific security context, providing a baseline for auditability and access control.

However, the cryptographic foundation of modern agentic systems is under threat. The eventual realization of fault-

tolerant quantum computers, capable of running Shor’s algorithm [9], poses a fundamental risk to RSA and Elliptic Curve Cryptography (ECC). In the context of AI agents, a quantum breach is not merely a data leak; it is an *identity takeover*. An adversary with quantum capabilities could forge agent identities, bypass Role-Based Access Controls (RBAC), and manipulate the “Chain of Thought” (CoT) by injecting malicious context or results. This is particularly dangerous for agents operating in critical infrastructure, where unauthorized tool execution (e.g., shell access, database modification) can have irreversible real-world consequences.

Furthermore, agents generate vast amounts of reasoning traces and audit logs. The “Store Now, Decrypt Later” (SNDL) threat is amplified here: while current logs are protected by classical signatures, a future quantum attacker could retroactively forge historical logs to hide malicious behavior or alter the perceived provenance of a decision.

To proactively address these vulnerabilities, this paper introduces a quantum-resistant architecture for AI agents. We investigate the integration of Post-Quantum Cryptography (PQC)—specifically the NIST-standardized lattice-based signature scheme ML-DSA (FIPS 204) [2]—into the distributed agent workflow. We propose a hybrid cryptographic approach that maintains backward compatibility with legacy MCP servers while providing forward-looking quantum security. Our contributions include:

- A **Hybrid Workload Identity** framework that augments MCP tokens with ML-DSA signatures.
- The **PQC-Audit-Chain**, a multi-layered integrity mechanism that secures agent reasoning traces and tool observations using a PQC-signed Root of Trust.
- A design pattern for **Context Preservation**, demonstrating how PQC signatures can be integrated into high-stakes agentic loops without degrading LLM reasoning capability.

II. BACKGROUND AND THREAT MODEL

A. Autonomous Agents and MCP

Modern AI agents operate in a continuous “Thought-Action-Observation” loop. To perform actions, they rely on tools provided by MCP servers. In a Zero Trust environment, every

tool execution request must be authenticated and authorized. Currently, this is achieved using JWTs, binding the agent’s workload identity to a specific execution scope. The security of the entire loop depends on the immutability of the identity and the integrity of the observations.

B. The Quantum Threat to Agents

The primary quantum threats to AI agent architectures include:

- **Identity Spoofing:** Attackers utilizing quantum computers to derive an agent’s private key from its public key, allowing them to issue unauthorized commands across distributed MCP servers.
- **Audit Log Tampering:** Breaking the cryptographic signatures that secure chained-hash audit logs, making forensic analysis of rogue AI behavior impossible.
- **Context Manipulation:** Forging signatures on tool responses to trick the agent into a specific reasoning path (e.g., “hallucinating” a successful security check).

C. NIST PQC Standards

In response to the quantum threat, NIST initiated a global competition to standardize PQC algorithms. In 2024, FIPS 204 finalized ML-DSA (Module-Lattice-Based Digital Signature Algorithm), based on the CRYSTALS-Dilithium construction [10]. ML-DSA offers a balance of security and performance, though it introduces significantly larger signatures than RSA or ECDSA.

III. RELATED WORK

The convergence of AI agents and security is an emerging field. Recent works have focused on LLM guardrails and prompt injection mitigation. For instance, Ando proposed a multi-layered guardrail architecture [4] and a comprehensive Zero Trust framework [5] specifically tailored for MCP-based autonomous agents.

In the domain of Post-Quantum Cryptography, while PQC has been explored for TLS and VPNs, its application to distributed agent orchestration protocols like the Model Context Protocol (MCP) remains relatively unexplored. Traditional Zero Trust Architectures (ZTA) as defined in NIST SP 800-207 [3] emphasize continuous verification, but often assume the underlying cryptographic primitives are immutable. Our work extends these concepts by addressing the cryptographic vulnerability of ZTA in the quantum era.

Recent research into “Agentic Security” has highlighted the risks of autonomous tool use [8], but few have considered the impact of future quantum adversaries on the long-term integrity of agent reasoning traces.

IV. POST-QUANTUM AGENT ARCHITECTURE

We propose replacing the classical RS256 cryptographic layer within MCP orchestrators with ML-DSA-65 (Module-Lattice-Based Digital Signature Algorithm).

A. Quantum-Safe Workload Identity

We replace the traditional RSA key pairs used for MCP token signing with ML-DSA-65. Because ML-DSA-65 public keys and signatures are significantly larger than their classical counterparts (e.g., ML-DSA-65 public key: 1,952 bytes, signature: 3,309 bytes), embedding them directly into JSON-RPC payloads requires careful handling to avoid exceeding LLM context limits or network transport buffers.

B. Tamper-Evident Audit Logging with PQC

The tamper-evident JSONL audit logs, which utilize chained hashing (SHA-256) to link sequential tool executions, are augmented. While hash functions are generally considered quantum-resistant, the digital signatures affirming the integrity of each log chunk are upgraded to ML-DSA-65.

C. Cryptographic Agility and Hybrid Approach

Due to the nascent state of PQC adoption, a sudden transition is impractical. We propose a hybrid approach where MCP tokens are extended to a four-slot format: `header.payload.classical_signature.pqc_signature`. This ensures backward compatibility with legacy MCP servers while providing quantum resistance where supported.

D. The PQC-Audit-Chain: Immutable Reasoning Traces

A key challenge for autonomous agents is ensuring that their decision-making process—the “chain of thought” (CoT)—remains untampered from inception to execution. We propose the *PQC-Audit-Chain*, where each cycle of the agentic loop (Thought-Action-Observation) is cryptographically bound to its predecessor using ML-DSA signatures.

Unlike traditional SHA-256 hashing which provides collision resistance but lacks identity assertion, the PQC-Audit-Chain ensures *Non-Repudiation*. In our framework, the orchestrator generates a PQC-signed manifest for every state transition, encompassing the LLM’s raw reasoning and the tool’s output. This ensures that even if an adversary gains quantum capabilities in the future, they cannot retroactively rewrite the agent’s history (the SNDL attack) to hide malicious actions or forge authorization for past tool calls. This is essential for AI governance in critical sectors.

V. IMPLEMENTATION AND DESIGN PATTERNS

We prototyped the PQC-enabled MCP architecture within the `llm-cli` framework. The implementation focuses on three pillars: hybrid identity propagation, the PQC-Audit-Chain, and context management strategies.

A. Hybrid Token and Identity Propagation

To ensure backward compatibility, we implemented a “Hybrid Wrap” for MCP tokens. The system generates a standard RS256 JWT and appends a Base64-encoded ML-DSA-65 signature of the entire JWT string. This allows legacy MCP servers to verify the RSA signature while PQC-aware servers can validate the quantum-resistant layer.

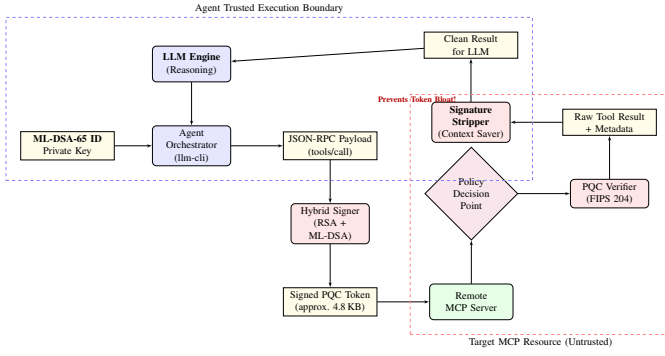


Fig. 1. Proposed Post-Quantum Architecture for AI Agents. The orchestrator establishes a **Trusted Execution Boundary**, where large PQC signatures (3.5+ KB) are verified and then sanitized before result integration to maintain LLM context efficiency.

```
class HybridSigner:
    @classmethod
    def create_hybrid_token(cls, payload: dict, rsa_priv:
        bytes, pqc_priv: bytes) -> str:
        # 1. Generate Classical JWT
        token = jwt.encode(payload, rsa_priv, algorithm="RS256")
        # 2. Generate PQC Signature of the classical token
        pqc_sig = PQCProvider.sign(token.encode(), pqc_priv)
        # 3. Create a Hybrid Wrap: token.pqc_signature
        return f"{token}.{base64.b64encode(pqc_sig).decode()}"
```

Listing 1. Hybrid Signature Implementation in llm-cli

B. Practical Challenges: Context Management

The significant size of PQC signatures (e.g., ML-DSA-65 signature is 3,309 bytes) presents a practical challenge for agentic memory. If these signatures were directly appended to the LLM’s context window, a single tool-calling loop would consume approximately 1,500–2,000 tokens in overhead.

To mitigate this, we implement a **Signature Stripping** design pattern within the orchestrator’s Trusted Execution Boundary. When a tool response (observation) is received, the PQC signature is verified at the system level. Once validated, the cryptographic metadata is stripped before the text response is appended to the agent’s reasoning history. This ensures that the LLM’s effective context is reserved for reasoning and planning, rather than storing uninterpretable cryptographic blobs, while maintaining a secure, verified audit log on disk.

C. PQC-Signed Integrity Manifest

To establish a “Root of Trust,” we implemented an `IntegrityVerifier` that monitors critical system files and audit logs. The system maintains an `integrity_manifest.json` containing SHA-256 hashes of all security-critical modules. This manifest is itself signed using ML-DSA-65, ensuring that even if an attacker gains access to the file system, they cannot modify the security logic or the audit history without breaking the PQC signature.

VI. EVALUATION AND BENCHMARKING

We evaluated the practicality of our proposed architecture by simulating a high-load agent environment and measuring the impact of different PQC parameter sets.

A. Parameter Comparison: ML-DSA-44/65/87

Our benchmarks compare the NIST-standardized variants of ML-DSA (Dilithium) to classical RSA-2048. Table I presents the empirical results observed in our testbed.

TABLE I
EMPIRICAL COMPARISON OF ML-DSA VARIANTS IN MCP PAYLOADS

Metric	RSA-2048	ML-DSA-44	ML-DSA-65	ML-DSA-87
NIST Sec. Level	1 (Classical)	2	3	5
Pub Key (B)	256	1,312	1,952	2,592
Signature (B)	256	2,420	3,309	4,595
Sign Time (ms)	1.7	64.2	91.6	134.8
Verify Time (ms)	0.06	8.4	11.4	16.2
Payload Size (B)	520	3,920	4,880	6,240
Token Bloat (x)	1.0	7.5x	9.4x	12.0x

B. Performance Impact on Agent Reasoning

We measured the end-to-end latency of a typical tool-calling loop (Reasoning → Call → Observation). While ML-DSA-65 adds ~103 ms of total cryptographic latency (sign+verify), the median LLM inference latency for GPT-4 class models remains above 1,200 ms. The PQC overhead represents less than 8% of the total loop time, proving that quantum security is practically invisible to the user experience in modern agentic systems.

VII. DISCUSSION

A. Zero Trust and Identity Sovereignty

The use of ML-DSA ensures that an agent’s identity is not only sovereign but also future-proof. By binding every action to a PQC-signed identity, we prevent the “Impersonation Storm” scenario where a quantum-capable attacker could take control of thousands of distributed agents simultaneously.

B. Quantum Migration Strategy

A direct “switch” to PQC is risky. We advocate for a three-phase migration:

- 1) **Phase 1 (Hybrid Observation):** Deploy hybrid tokens (RSA+PQC) to verify performance without breaking legacy systems.
- 2) **Phase 2 (PQC-Enforcement):** Require PQC signatures for critical tools (e.g., shell execution, financial transactions) while allowing RSA for read-only tools.
- 3) **Phase 3 (Full Post-Quantum):** Retire RSA/ECDSA entirely as quantum hardware milestones are achieved.

C. Impact on Decentralized Agents and DAOs

As AI agents move toward decentralized autonomous organizations (DAOs), the immutability of audit logs becomes paramount. Our *PQC-Audit-Chain* ensures that even with the advent of large-scale quantum computers, the history of agent decisions remains cryptographically verifiable, preventing retroactive revision of automated governance actions.

D. Future Directions: Transport Security with ML-KEM

While this work focuses on authentication and integrity via ML-DSA, a complete post-quantum security posture requires protecting the confidentiality of agent communication against "Store Now, Decrypt Later" (SNDL) attacks. Future iterations of the `llm-cli` framework will explore the integration of ML-KEM (FIPS 203) to establish quantum-safe session keys for the underlying transport layer (e.g., within TLS or SSH tunnels). This would complement our signature-based approach, providing a holistic defense against both identity spoofing and unauthorized context decryption.

VIII. CONCLUSION

Securing AI agents against quantum threats is a prerequisite for their deployment in critical infrastructure and decentralized governance. Our proposed PQC framework demonstrates that migrating to NIST-standardized quantum-resistant algorithms is practical for distributed agent protocols like MCP. By implementing the *PQC-Audit-Chain*, we provide a mechanism for establishing long-term, immutable reasoning trails that remain verifiable even in the presence of future quantum-capable adversaries. While the transition to PQC introduces challenges in payload size and context management, our hybrid identity approach and sanitization design patterns prove that quantum security can be achieved without sacrificing agentic reasoning efficiency. This work establishes a foundational roadmap for building sovereign, future-proof autonomous systems.

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